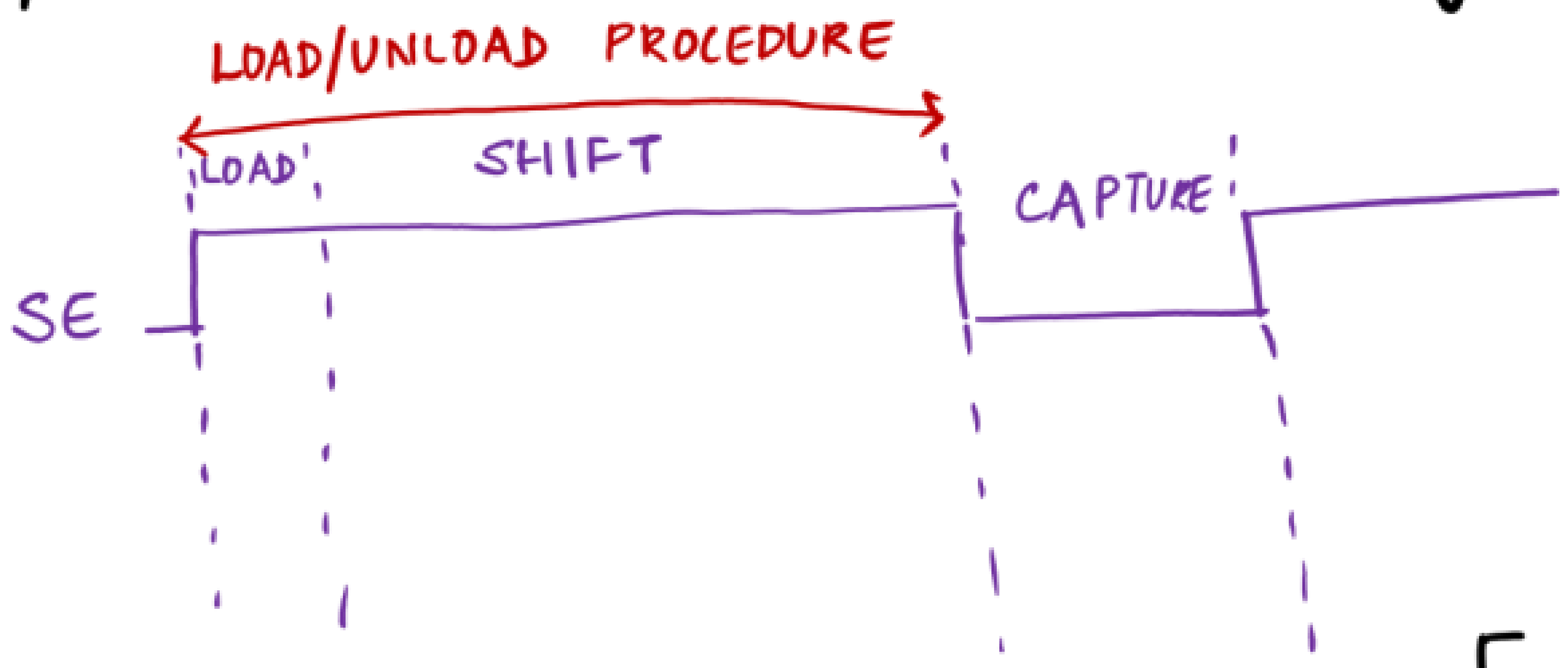
Level 4 projects

Part 2

ATPG :-

DRC :- DI violation

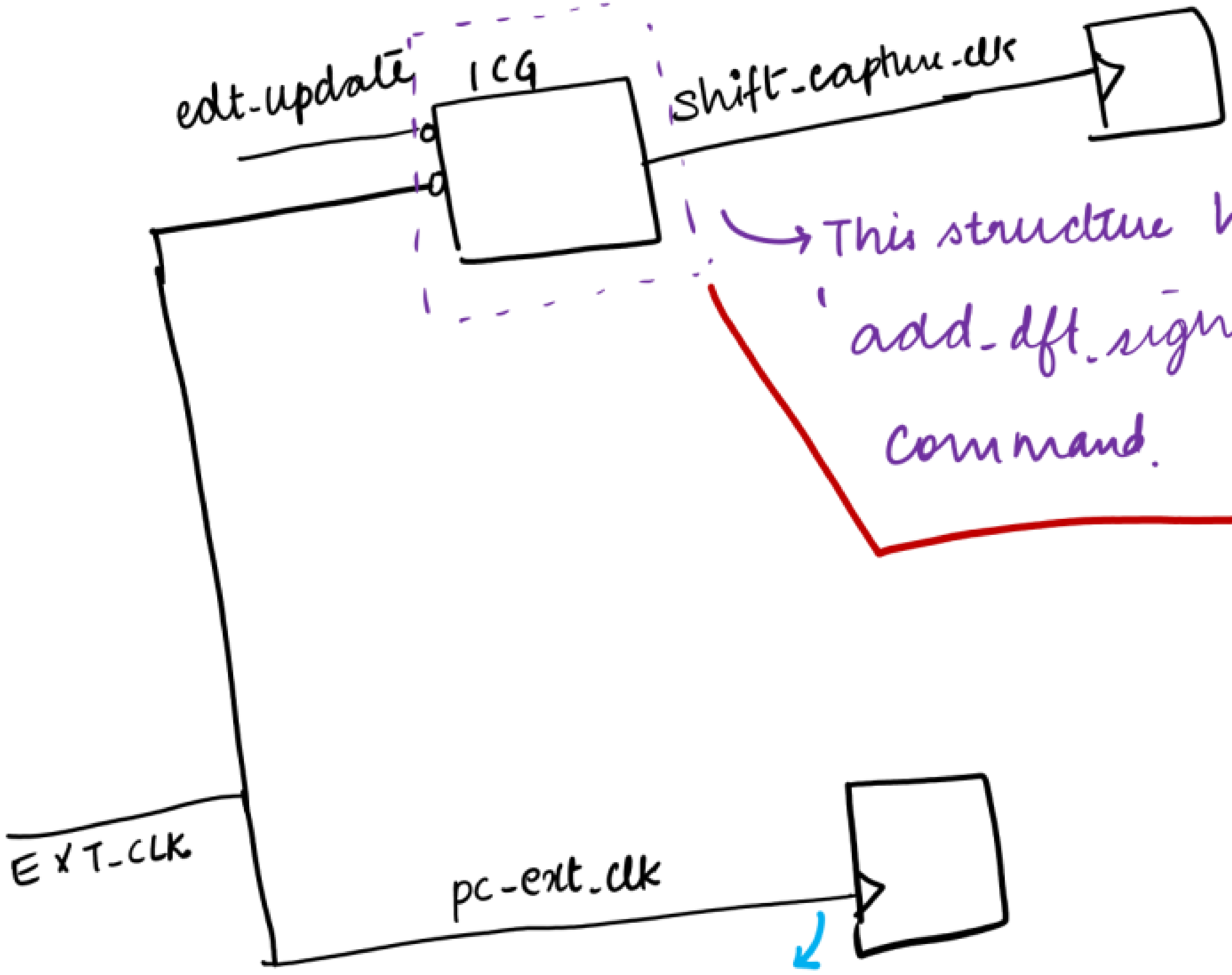
<flop name> is disturbed during load/unload procedure.



CAUSE :- if a scan clk is pulsed during load phase, it will cause DI violation.

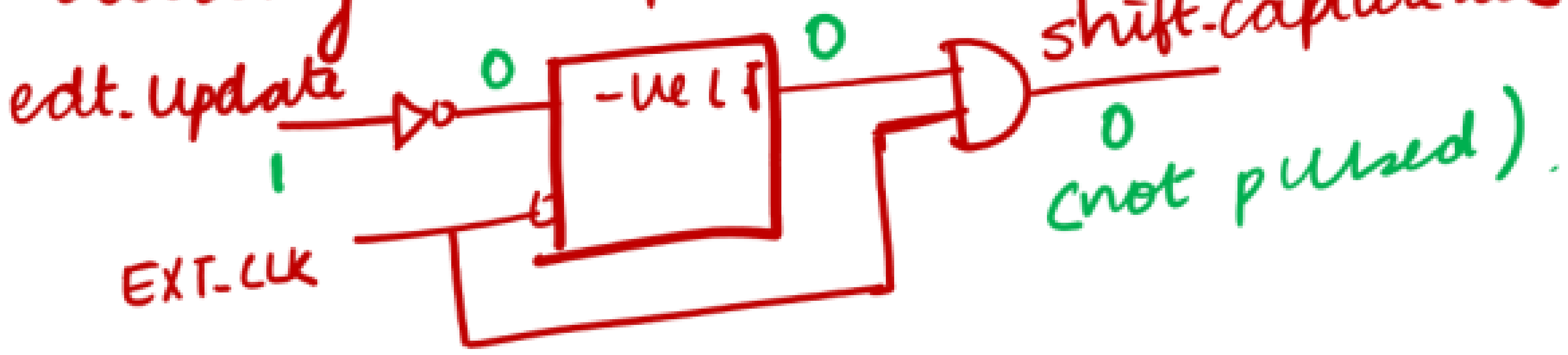
[During load phase, only edt_clock and edt_update has to be pulsed]

Scan clock should not be pulsed during the load phase.



This structure has got inserted when we have given the 'add_dft_signal shift_capture_clk -create_from_other_signals' command.

This structure will not allow scan clk to be pulsed during load phase.

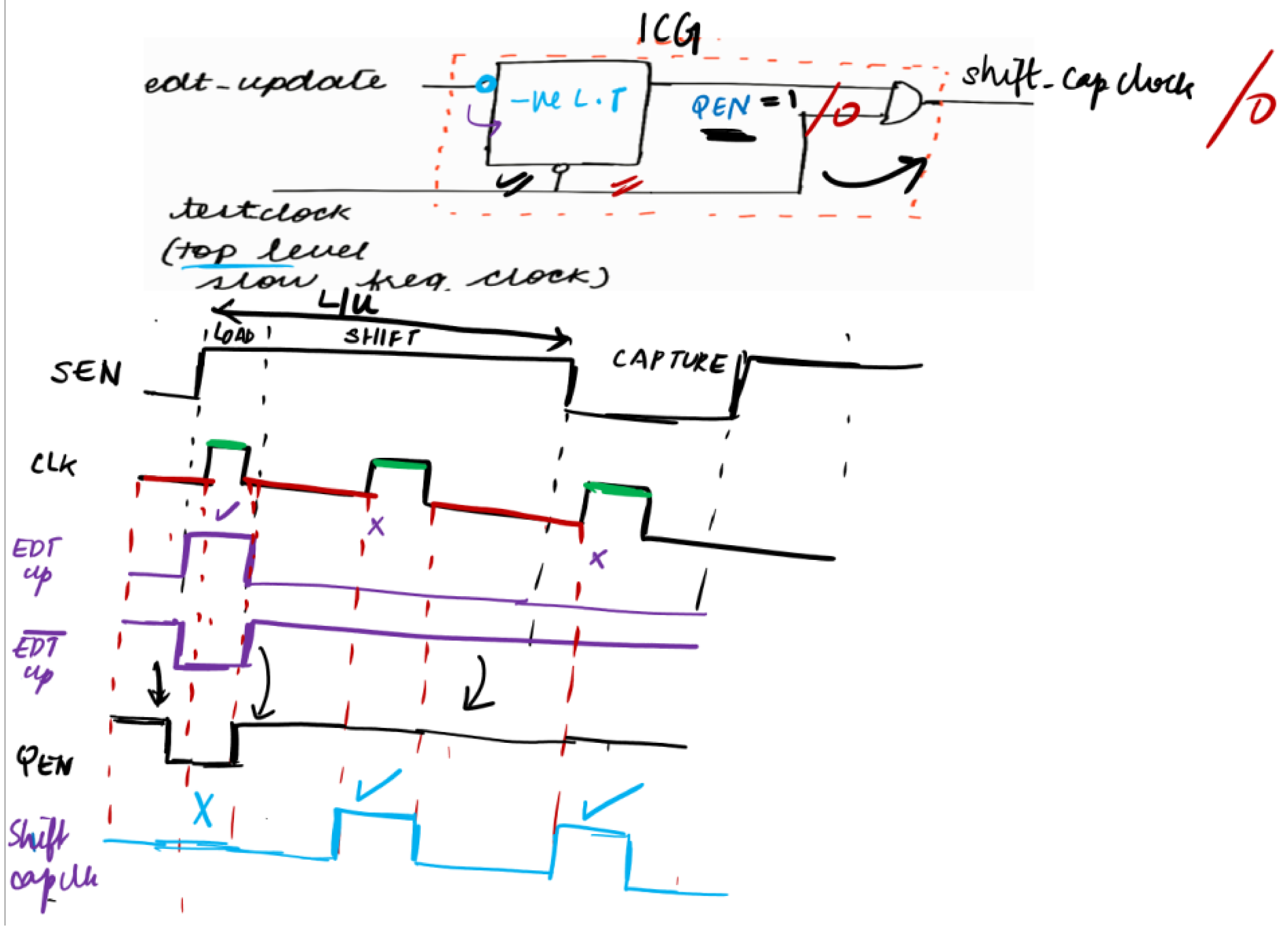


During load phase :

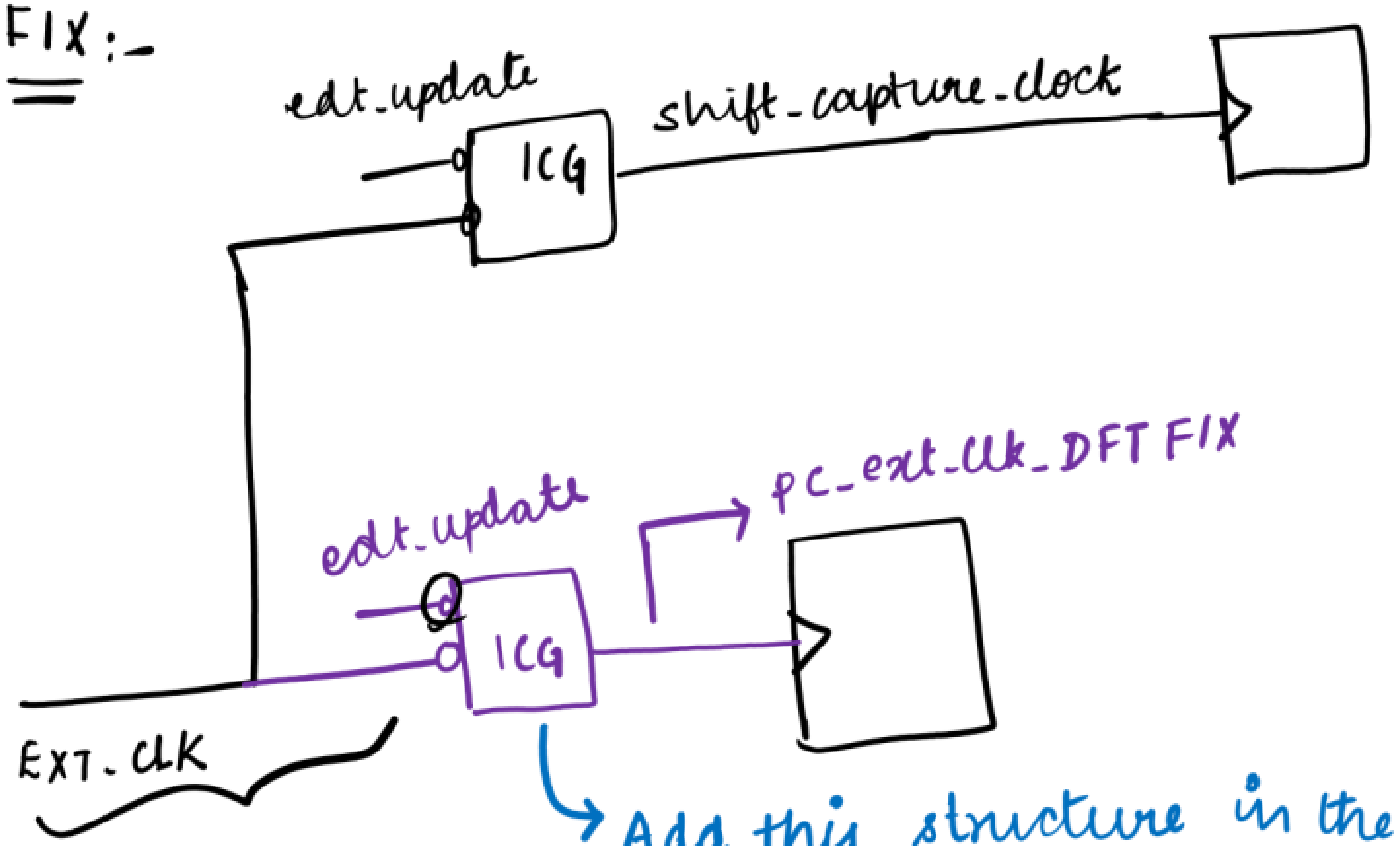
See next page for waveform explanation.

for this flop, EXT_CLK is directly connected. So for this flop, it's scan clk is pulsed during the load phase.

∴ We get D1 violation.


ICG
edt_update
-ve L.T
QEN = 1/0
shift_cap clock
1/0
testclock
(top level
slow freq. clock)
4U
LOAD
SHIFT
CAPTURE
SEN
CLK
EDT up
EDT up
QEN
Shift cap clk

FIX:-



Add this structure in the scan inserted netlist.

ATPG coverage analysis :-

coverage → 91 %

Reason :- .tcd-mem_lib file of memories

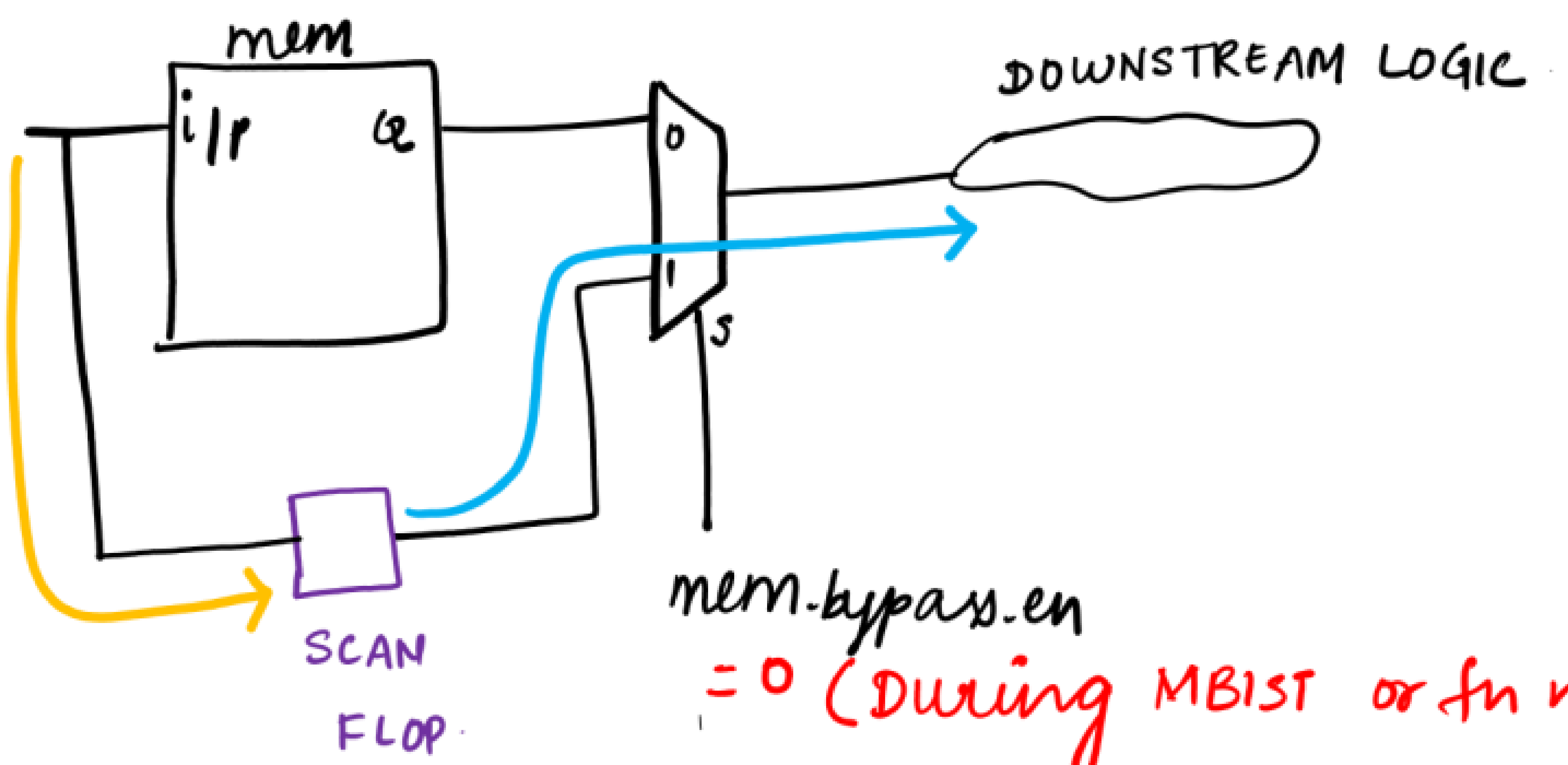
Transparent Mode : None

↳ Memory bypass logic will not be added by the tool. So the memories are not bypassed during ATPG time.

Fix :- Transparent Mode : syncMux

↳ Tool will automatically add the logic to bypass the mem. So memories are bypassed during ATPG.

∴ cov = 96.71 %



= 0 (During MBIST or fn mode)

= 1 (otherwise)

SCAN FLOP

☐ :- * It will act as an obs pt for the fault sites at the i/p of the memory.
(i.e; the value at the i/p of mem is captured into that flop).
[Depicted by yellow colour line]

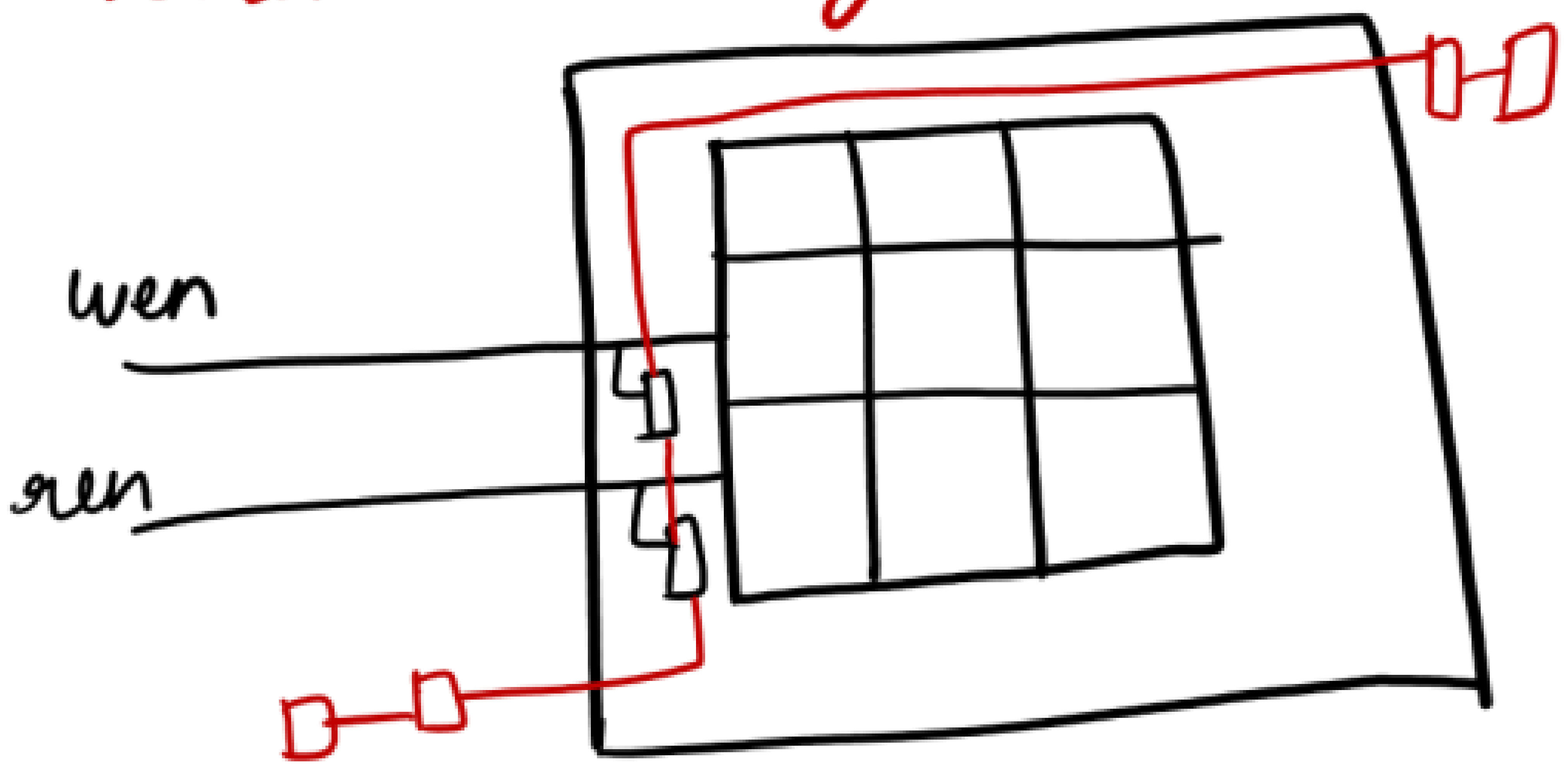
* It will act as a control flop for the downstream logic.
(i.e; the last shifted value of that flop will control the downstream logic)
[Depicted by blue colour line]

NOTE:- The value only at the Data i/p of the memory will be observed after adding the above logic.

But the value of chip select (CS), RE, WE, ADDR is not observed.

In industries, observation logic for WE, RE etc will be inside the memory and it will be supplied by the memory vendor.

∴ The faults in those lines will get observed.



If the faults in these lines get detected, then our coverage would increase from 96.71 % to 99 %

But the mem which is given in L4 projects does not have this support.

$\therefore$ We are not able to detect the faults in those lines.

$\therefore$ Our cov $\simeq$ 97 %

If you want to detect the faults in those lines, we have to manually add the obs pt outside the memory and stitch it into scan chains.

MBIST simulation failures in L4 projects :-

CAUSE1 :- if we have not read the RTL files of memories (.v files), then it will fail.

Fix :- Find all the .v files of memories and re-direct the o/p to a file

⇓

This is already done and kept in a file named as new_modules.dofile and it is read during MBIST insertion

∴ We will not face any problem.

CAUSE 2 :-

In questa_simulation script, tool will be reading all power library files.

As a result, all the cells in the design will be considered as power cells.

The VDD and VSS ports of those cells will be undriven. So the o/p of the cells will be X.

Hence we will get mismatches

Fix :- Comment off all the lines in the questa-simulation.script where the tool is reading the power lib file.

1. MBIST_insertion/simulation_outdir/cpu-top-cpu_top_mbist_sim_signoff/memoryBist-P1_sim/

questa_sim_script

file where we need to make the change.

NOTE :- Diff b/w power lib and normal lib

In normal lib, VDD and VSS ports of the cells will not be mentioned

In power lib, it will be mentioned

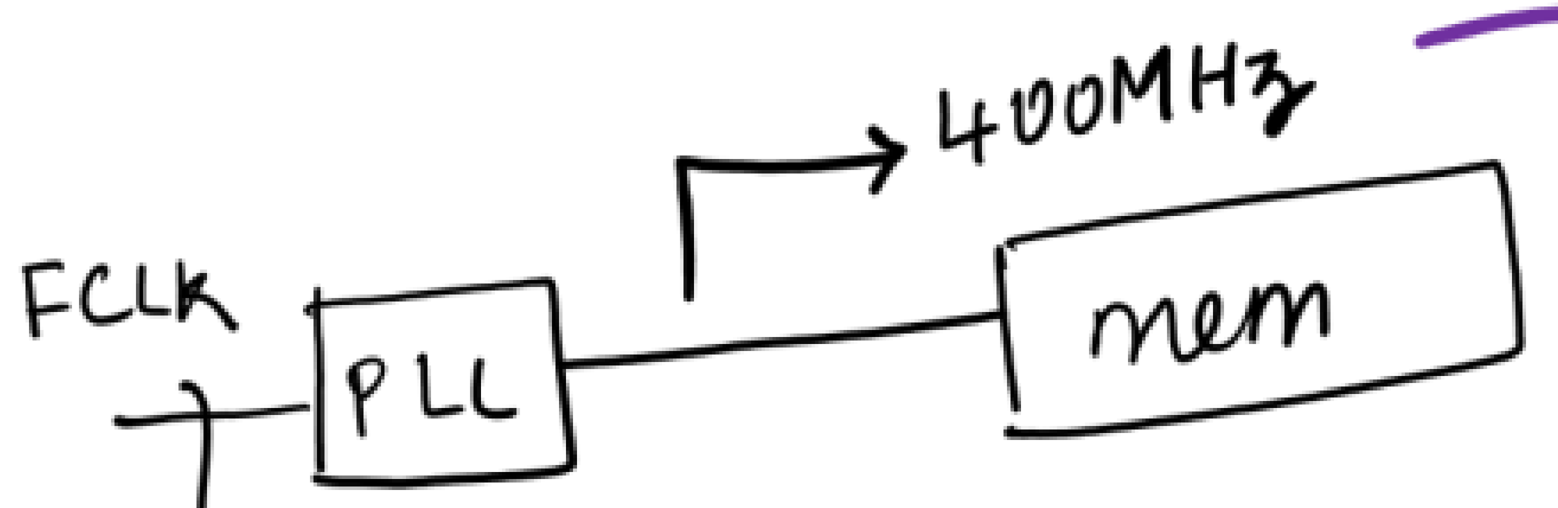
The power.lib file has to be read only when we have pg (power ground) verilog netlist.

CAUSE 3 :-

clock monitoring failed

Period expected = 2.5ns

Actual = n/a (no transition detected)



$$T = \frac{1}{f} = \frac{1}{400 \times 10^{6}}\, s = 0.0025 \times 10^{-6}\, s$$

$$= 0.0025 \times 10^{-6} \times \frac{10^{-3}}{10^{-3}}\, s$$

$$= \frac{0.0025\, ns}{10^{-3}} = 2.5\, ns$$

$$\boxed{T = 2.5ns}$$

Coming from crystal oscillator which can produce only around 60MHz clock. So PLL will be used as freq multiplier

We have used an internal clock.

∴ QuestaSim tool will not understand that.

So it will throw the clock monitoring failed error.

Fix :- Force the clock in the testbench

[MemoryBistP1.v]

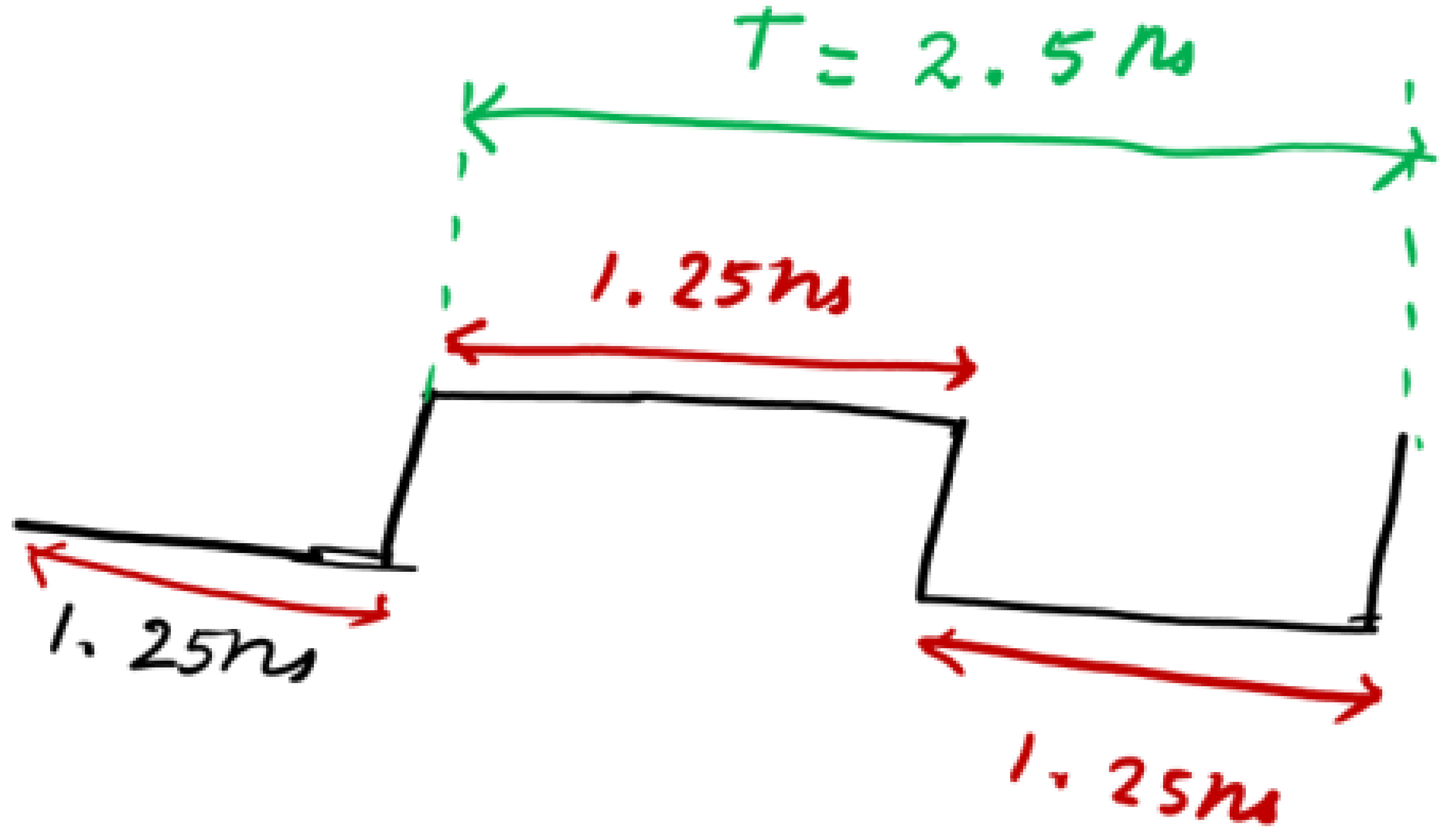
{in tsdb_outdir/patterns/cpu-top_cpu-top_mbist_patterns_signoff}

REFERENCE FILE :- Keep this file as a reference while editing in your original testbench.

1. MBIST_insertion/MemoryBist_P1.v

↳ This is the fixed file

```
initial begin
      force DUT_inst.PLL.CLKOUT = 1'b0;
   #0;
    forever begin
     #1.25;
        force DUT_inst.PLL.CLKOUT = 1'b1;
     #1.25;
       force DUT_inst.PLL.CLKOUT = 1'b0;
end
end
```



CAUSE 4 :-

clock monitoring failed

period expected = 2.5ns

actual period = 2.6ns

Reason :- QuestaSim tool rounds off 1.25 as 1.3

1.25 + 1.25 = 2.5 (expected period)

↳ 1.3 + 1.3 = 2.6

⇓ becomes the simulation clock period

Fix :- In the testbench file → MemoryBistP1.v

{ in tsdb-outdir/patterns/cpu-top cpu-top_mbist_patterns_signoff }

change the precision in timescale.

`timescale 1ns/100ps ⇒ original

`timescale 1ns/10ps ⇒ fix.

NOTE:- PLLs will be programmed through JTAG while writing patterns for tester (STIL pattern) to generate the internal clock.

There will be a guide in each company to program the TDR.

↳ The ref clock will be coming from crystal oscillator.

using the ref-clock, the PLL has to be programmed (VCO values, div/mul value etc has to be programmed) to get the req freq at the O/P of PLL.

This programming is done using JTAG while generating patterns for tester (STIL pattern). These values will be specified by the designer.